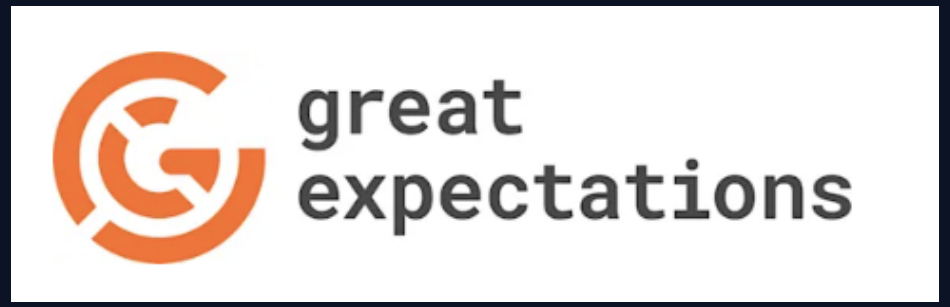
Great Expectations

Data Quality Framework

Reliable data before model training



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github.com/ibrahim-bancar/great-expectations-mlops

WHY DATA QUALITY?

Garbage in, garbage out.



Loss: $12.9M

Source: Gartner research (2020)





https://www.gartner.com/en/data-analytics/topics/data-quality

Medial appointment data

PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	Diabetes	Alcoholism	Handcap	SMS_received	No-show
29872499824296		F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	0	0	0	0	No
558997776694438	5642549	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	0	0	0	0	No
4262962299951	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	-5	MATA DA PRAIA	0	0	0	0	0	0	No
867951213174	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	1000	PONTAL DE CAMBURI	0	0	0	0	0	0	No
8841186448183	5642494	X	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	1	0	0	0	No
95985133231274	5626772	F	2016-04-27T08:36:51Z	2016-04-29T00:00:00Z	76	REPÚBLICA	2	1	0	0	0	0	No
733688164476661	5630279	F	2016-04-27T15:05:12Z	2016-04-29T00:00:00Z	23	GOIABEIRAS	0	0	0	0	0	0	Maybe

Source: https://www.kaggle.com/datasets/joniarroba/noshowappointments

Medial appointment data

Missing value

Invalid age (negative)

Invalid age (unrealistic)

PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	Diabetes	Alcoholism	Handcap	SMS_received	No-show
29872499824296	—	F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	0	0	0	0	No
558997776694438	5642549	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	0	0	0	0	No
4262962299551	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	-5	MATA DA PRAIA	0	0	0	0	0	0	No
867951213174	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	1000	PONTAL DE CAMBURI	0	0	0	0	0	0	No
8841186448183	5642494	X	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	1	0	0	0	No
95985133331274	5626772	F	2016-04-27T08:36:51Z	2016-04-29T00:00:00Z	76	REPÚBLICA	2	1	0	0	0	0	No
733688164476661	5630279	F	2016-04-27T15:05:12Z	2016-04-29T00:00:00Z	23	GOIABEIRAS	0	0	0	0	0	0	Maybe

Invalid category (unexpected)

Invalid value (out of range)

Source: https://www.kaggle.com/datasets/joniarroba/noshowappointments

The GX validation workflow



Defining Expectations

We define data quality rules (expectations) for the medical appointment dataset.

Individual Expectations

AppointmentID

Values must not be null.

Example invalid value:

—

Gender

Values must be in the set {F, M}.

Example invalid value:

X

Age

Values must be between 0 and 120.

Example invalid value:

-5, 1000

Scholarship

Values must be in the set {0, 1}.

Example invalid value:

2

Expectation Suite

medical_appointments_suite

- AppointmentID not null
- Gender in {F, M}
- Age between 0 and 120
- Scholarship in {0, 1}

Running Validations

Data (Batch)

AppointmentID	Gender	Age	Scholarship
5642903	F	62	0
5642503	M	56	1
—	X	-5	0
...	...	...	...

Expectation Suite

- AppointmentID not null
- Gender in {F, M}
- Age between 0 and 120
- Scholarship in {0, 1}

medical_appointments_suite

Validation Definition

Batch
(medical_appointments)

+

Expectation Suite
(medical_appointments_suite)

medical_validation

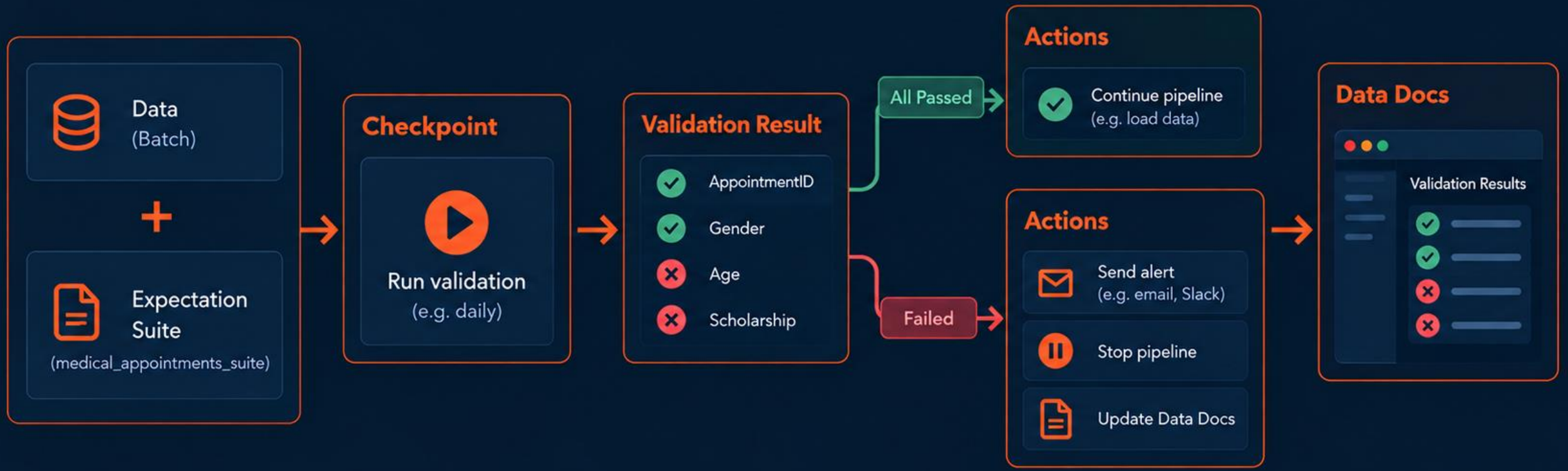
Validation Results

FAILED

Expectation	Result	Details
AppointmentID not null	✕ FAIL	1 null value
Gender in {F, M}	✕ FAIL	1 unexpected value (X)
Age between 0 and 120	✕ FAIL	2 unexpected values (-5, 1000)
Scholarship in {0, 1}	✕ FAIL	1 unexpected value (2)

Checkpoint, Actions and Data Docs

A checkpoint runs validation(s) and triggers actions based on the results.



Limitations of Great Expectations

1

Manual Rule Definition

Users must define what valid data looks like.



2

No Automatic Cleaning

Great Expectations detects issues but does not fix them.



3

Performance Overhead

Validations add extra computation, especially on large datasets.



Thank You!

Great Expectations helps us build more reliable data pipelines by catching data quality issues early.



Questions?

Reference: Great Expectations Official Website https://greatexpectations.io/